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An ML-driven decision support system for engineering college selection post-intermediate education in India

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Abstract

The paper is a developed machine learning-based decision support system that will guide the choice of an engineering college by intermediate (10+2) students in India in accordance with their academic backgrounds, individual preferences and institutional characteristics. The model combines information that is heterogeneous, such as entrance exam scores, college ratings, placement scores, accreditation, tuition costs, geographic orientation and alumni responses to create customized college suggestions. Decision Tree, Random Forest, Support Vector Machine (SVM), k-Nearest Neighbours (KNN) and Logistic Regression. Five machine learning algorithms were trained and tested to determine which predictive model is most efficient. The best of them was the Random Forest algorithm, which obtained the highest accuracy, with 87.3, and an F1-score of 0.85, showing the best generalization and strengths. Furthermore, a hybrid recommendation scheme that incorporates both content-based and collaborative filtering strategies was introduced to increase the personalization and to adequately address the issue of cold-start. The system suggested can be scaled to support the large-scale implementation and provide scalability, interpretability, and flexibility in a wide range of educational settings. The results of the experiments prove that the hybrid framework has greater precision and recall than standalone recommendation models. This study fills an important gap in automated educational guidance by offering a transparent, unbiased and data-driven decision support system to the student, counsellors and policy makers. The created architecture represents a groundbreaking move to incorporate artificial intelligence and predictive analytics in the educational planning, which will make the process of college selection informed, equitable, and evidence-based and will make college selection available to aspiring engineering students in India.

Keywords Machine learning, Educational recommendation systems, Engineering college selection, Predictive analytics, Decision support systems, SDG4

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1 Introduction

The choice of a good college of engineering following an intermediate education is one of the most important choices in the academic life of a student. Indian engineering colleges are more than 4000, and the choices available to students and their families are many, which can be very confusing, especially in terms of becoming informed about them. Outdated systems of deciding on college are grounded on incomplete information, peer recommendation, as well as minimal information, and hence make suboptimal choices that can have a significant impact on career choices. There are several issues associated with the modern world:

1. Lack of information between students and learning institutions.
2. The absence of individual guidance.
3. Manual research is time-consuming.
4. Lack of consistency in the standards of evaluation of the various sources of information.
5. Poor combination of various decision factors.

1.1 Significance of automating college selection using ML

The solutions of machine learning can have a radical potential to overcome such obstacles by:

6. Personalization: ML models may be individual ones that are analyzed based on the profile of a particular student and his/her likes and dislikes to provide personal recommendations.
7. Flexible analysis: Algorithms can handle large volumes of data in real-time and on varied data.
8. Predictive capabilities: ML models can make predictions of whether students will be successful in various institutions.
9. Scalability: Thousands of students can be served by the automated systems at the same time.
10. Continuous learning: Models can enhance recommendations based on past results and comments.

1.2 Objectives of the study

The main aims in this study are:

1. To develop a more holistic machine learning system to create a college recommendation engine.
2. To evaluate the performance of different Machine Learning algorithms in the prediction of an appropriate college match.
3. To apply and test various approaches to recommendation systems.
4. To develop a scalable system that can be deployed with existing educational systems.
5. To mitigate ethical issues such as bias and ethical algorithms.

1.3 Contributions of the study

To deal with the multidimensional issues associated with the choice of an engineering college following the intermediate education, the study offers several relevant contributions:

1. A unified ML-based recommendation framework: An overall decision-support framework that integrates classification models, hybrid recommendation methods and domain specific filters used to make the recommendation framework in line with Indian engineering education environment is developed.
2. Strict comparative analysis of machine learning algorithms: Five types of algorithms were already investigated and compared Decision Tree, Random Forest, SVM, KNN, and Logistic Regression with the highest accuracy (87.3%), F1-score (0.85) was achieved by the random forest.
3. Hybrid recommendation engine: A hybrid content-based and collaborative filtering was presented, which overcome cold-start problems and performed better based on MAE, RMSE, Precision and K and Recall and K and NDCG.
4. Another research has introduced an advanced feature engineering and composite indicators where new indexes like: Affordability Index, Career Alignment Score, Location Convenience Score, Academic Preparedness Score, etc. were introduced and they improved predictive strength and interpretability.
5. Ethical and transparent AI framework: prevention of bias thru strategies, prohibition of unfairness thru restrictive mechanisms (e.g. feature importance analysis), and interpretable mechanisms sustain ethical and equitable suggestions.
6. Scalable system architecture: It uses modular and cloud-scaled components alongside caching, approximate nearest neighbour search, and efficient similarity computation to be implemented to enable the deployment at scale.
7. Early user tests: A real-user study involving 50 intermediate students demonstrated that the system was quite useful (92%), relevant (88%), and satisfied (4.4/5) indicating the reality of the usefulness of the system.

1.4 Background

The engineering education system in India is one of the most diverse and largest education systems in the world that admits millions of students in over 4000 engineering institutions. After the completion of intermediate (10+2) education, students must make a fateful and sometimes irreversible choice on college selection, at a very young age in career development. The long-term effects of this decision are on academic performance, employability, socio-economic mobility, and job satisfaction [17, 18].

Historically, in India, the choice of the engineering college has been influenced by a few factors including the rank in the entrance examination, the reputation of the college, peer pressure and informal guidance. As much as national-level entrance exams and uniform ranking systems like the National Institutional Ranking Framework (NIRF) offer objective indicators about how institutions can be compared, multidimensional compatibility between an institution and a student is not completely represented [27]. Such issues as financial affordability, geographic accessibility, and the availability of programs, placement outcomes, accreditation status, and personal preferences of students are usually either viewed separately or not considered at all, which leads to fragmented and less than optimal decision-making.

This has been compounded by the fact that engineering institutions have grown at a rapid pace in the last twenty years. Despite the enhanced institutional capacity that has seen greater access to technical education, this has also created a significant amount of heterogeneity in the quality of academic provision, infrastructure, and faculty as well

as graduate employability outcome [32]. This inconsistency makes it hard to assess students and parents, especially with the first-generation learners and the students with less access to professional counselling and reliable institutional data .

Simultaneously, the availability of educational data continues to increase, and the performance of artificial intelligence has opened new prospects in the field of data-based educational decision support. Various datasets, including academic outcomes, entrance test scores, academic rankings, accreditation reports, fee schedules, placement rates, geographic characteristics, and student reviews, can now be using machine learning methods to be systematically analysed [8, 25]. Machine learning models can be considered especially useful in the context of these heterogeneous data, allowing them to reveal latent patterns and make personalized predictions on a large scale, which could not be achieved by using rule-based and heuristics-driven solutions.

In the wider area of educational technology, recommendation systems have been highly sought after in areas like course recommendations, personalization of learning resources and the prediction of academic performance [2, 7]. Nevertheless, relatively small body of research has been done on the holistic issue of the college selection, in Indian engineering education. Selection of colleges is a multi-criteria decision-making issue where there is a need to strike a balance between academic eligibility, financial viability, geographical limitations, quality of institutions and future career prospects [4, 43].

Combination of content-based and collaborative filtering to produce hybrid recommendation approaches have been demonstrated to be better than the standalone methods in enhancing personalization and addressing the cold-start issues, especially when dealing with new users with little interaction history [5, 12]. Simultaneously, predictive analytics has proven to be highly promising in predicting student-institution compatibility and future academic performance, hence improving the validity of recommendations [32].

Ethical issues also provide further reasons why responsible educational decision support systems are necessary. Previous research has been found to point out the dangers of algorithmic prejudice, absence of clarity, and inequity in treating students with diverse socio-economic or scholarly backgrounds [11, 42]. Therefore, fairness-conscious modelling, interpretable machine learning, and constraint-based filtering mechanisms have become critical design considerations to the contemporary educational recommendation systems [43].

It is on this background that the current research is located at the intersection of machine learning, educational data mining, and decision support systems. The proposed framework of suitability prediction via supervised learning and personalized ranking via a hybrid recommendation system aims to deliver a scalable, interpretable and ethically sound solution to the complicated problem of college selection engineering in India. This background provides the conceptual and practical background to the methodological design and empirical analysis that are to be conducted in the following sections of this manuscript.

1.5 Novelty of the present study

The current research is very different relative to other previous studies in the same field as they suggest a multi-dimensional, multi-layered ML based system of college selection engineering. As compared to the past research where an emphasis is put on course

recommendation or only on academic contributions, the current study combines a broader scope of variables such as financial, geographic, placement, accreditation and student-preference variables. It also proposes domain-specific composite measurements, like the Affordability Index and Career Alignment Score, incorporates a hybrid recommendation model to address cold-start problems, and implements fairness and explainability. Moreover, the system has scalable architecture and justified by real student feedback, it is more contextually strong and practically implementable than the previous methods.

1.6 Structure of the study

This study has been classified in six sections. Section 1, mentioned about the research problem, objective of the study, contribution and background of the study along with significance. Section 2 covers about the major literatures regarding the study, different theoretical variables, while section three covers about research methodology, sampling techniques, respondents' data operationalisation of the research. Section 4 have result and secondary data has been explored and interpreted with standard comparison of previous study. Section 5 covers discussion and finding of the study while section six covers the conclusion. Last section covers limitation and future scope of the study.

2 Literature review

2.1 Educational recommendation systems: foundations and approaches

The development of educational recommendation systems has been characterized by the shift towards a rule-based approach to more advanced machine learning models able to work with large and heterogeneous data. Older models were mainly concerned with the provision of static advice when considering very little user input, and the modern models are characterized by complex algorithms that are used in personalizing recommendations, predicting academic performance, and providing real-time advice provision to students. The increasing access to educational information, combined with the advances in artificial intelligence, has enabled scientists to create systems that can guide students in making complicated educational decisions.

One of the most popular methods is collaborative filtering, which is used to determine similarities between users or items based on patterns of interactions. The article by Kumar et al. [1] has employed user-based collaborative filtering in an e-learning setting and demonstrated that the latter can be used to establish trends within groups of students. They were able to tailor academic guidance by suggesting courses based on similarities in their behaviour, creating a basis of customization in academic guidance. It was later expanded by Bobadilla et al. [2], who have offered an in-depth review of the collaborative filtering techniques, enumerating many of the challenges, which often appear to be widespread among the collaborative filters, such as data sparsity and scalability, which nonetheless are indispensable in learning.

The content-based filtering systems, on the other hand, utilize the institutional properties and individual preferences to generate suggestions. Patel and Sharma [3] created a recommendation algorithm that relied on the attributes, including accreditation, the quality of the infrastructure, and the curriculum available to prescribe educational resources based on the profile of a student. Their contribution underlined the need to align the preferences of the students with the institutional advantages, but it

had limitations as no comprehensive evaluation systems and real-life validation were performed.

In acknowledgement of the shortcomings of single methods, a strong alternative has emerged as a hybrid recommendation system by using a mixture of collaborative and content-based methods. Singh et al. [4] introduced a hybrid university recommendation framework, which not only increased the diversity of recommendations but also addressed the cold start issue that plagues new users with a lack of historical interaction information. Formal taxonomy of hybrid methods has been previously given by Burke [5], who notes that combining several algorithms would add robustness to the system, and reduce overfitting and interpretability issues.

Other than methodology that relies on filtering, predictive analytics has become more relevant in the educational systems. [6] investigated the potential of historical academic data, entrance exam results, and institutional attributes to predict student achievement, thus providing the opportunity to take early interventions and offer personalized suggestions. Their study highlighted the radical change machine learning models can bring to the formation of student outcomes through predicting difficulties and offering guidance based on empirical evidence.

Recently, deep learning methods have extended the horizons of educational recommendation systems. Wang et al. [7] carried out a comprehensive literature review of the deep learning models used in educational data and addressed such models as recurrent neural networks, autoencoders, and neural collaborative filtering. These methods offer the capability to measure non-linear relationships involving student preferences and institutional characteristics that are hard to measure, but they are characterized by their interpretability difficulties that complicate counsellor-student relationships.

2.2 Feature engineering and data quality in educational systems

The success of the recommendation system can be defined by the quality/representation of data. Educational data sets are usually blemished with non-reporting instances, uneven reporting, and disparity in the type of data. Preprocessing and feature engineering of these issues are needed to ensure the accuracy and robustness of the models.

The survey of feature engineering methods that have been made to work with educational data is also quite comprehensive by Lee et al. [8]. Their work provided practical recommendations on the techniques of managing missing values by using statistical imputation methods such as mean substitution and the Multivariate Imputation by Chained Equations (MICE) algorithm. They also discussed ways of identifying outliers, like the Interquartile Range (IQR) method and Z-score, which will help remove bad data, and yet it will still have valid variability.

Also, biased distributions that are quite common in financial data, position data, and performance in academic institutions are typical examples of data where feature transformation techniques such as Min–Max scaling and logarithmic transformations are usually resorted to, to normalize the skewed data. Chen et al. [9] have proposed two composite scales, like the Affordability Index and Career Alignment Score, comprising a combination of several features that are more effective at predicting the financial capability and career interests of students. These capabilities that have been designed offer an informative base of recommendations on top of the raw data to add the nuances of real-world decision-making.

The other significant preprocessing action involves the reduction of dimensions due to dimensionality, because high-dimensional data that contains many correlated variables can cause noise and reduce the interpretability of the model. Lee et al. [10] emphasized the use of Principal Component Analysis (PCA) and the recursive feature elimination procedures to not only identify the most informative ones but also retain the semantic integrity of the data. This would imply that irrelevant features are not overloaded with models but focus on important features.

2.3 Ethical, fairness, and bias considerations

Even though the implementation of recommendation systems is expected to lead to increased personalization and efficiency, there are also certain ethical issues related to the system, in terms of bias, fairness, and privacy. The consequences of unfair recommendations could be dire in the educational context and could impact the academic and career paths of students.

Johnson and Smith [11] highlight in a radical contribution to the discussion about algorithmic prejudice in education that the proposals can be biased due to the existence of disparities in the socio-economic area, geographical limits, and unequal access to resources. Their findings showed that audits of fairness and limitations need to be included to ensure that recommendations do not place disadvantages on some groups of students in reverse. They further emphasized the issue of transparency, where recommendation models would clearly explain the factors that have led to their recommendations.

Zhang et al. [12] have addressed the cold start problem that affects users who have a low-interaction history, often that of underexplored communities. According to their research, they proposed hybrid techniques that integrate the domain knowledge and the collaborative cues, therefore improving the equity of the recommendations without compromising the accuracy.

The problem of privacy is also an elephant in the educational recommendation system, and sensitive information about students is often at play. Some of the excellent measures to anonymize, use encrypted protocols, and obtain user consent by Wilson and Davis [13] follow the data shield law, such as GDPR and the Digital Personal Data Protection Bill in India. Their recommendations report on the need to have ethical stewardship when managing personal information in scholarly institutions.

2.4 Personalization and context-aware recommendations

Most recently, the recommendation systems have attempted to transcend the academic performance of personalization to behavioural rules, psychological traits, and situational tastes. This growth is premised on the fact that the choices made by the students are guided by a multitude of factors.

Rodriguez and Thompson [14] touched on the idea of implementing personality tests within the context of the recommendation to make it more consistent with the coping styles of the students and their learning styles. Their contribution has brought it to light that behavioural data could be useful when it comes to improving student satisfaction and engagement.

To supplement this plan, Huang et al. [15] showed that identification of concealed emotional inclinations is possible using the sentiment analysis of student essays and

feedback. Their study allowed the systems of recommendations to know the subjective experiences of the students who apply the methods of Natural Language Processing (NLP) and make propositions that appeal to their interests and anxieties.

The time is also significant, as recommended by the educational recommendation systems. It is possible to correct the systems according to the evolving institutional performance and preference among students with the help of time-conscious models by attaching greater importance to the comparatively recent recordings as compared to the older ones. Lee et al. [16] revealed how the mechanism of temporal weighting may increase the relevancy of suggestions due to the factor of taking the trends of the academic performance and academic partnership into account.

2.5 Scalability and deployment challenges

Educational recommendation systems must be accurate, scalable, and efficient to be accepted on a large scale. Recommendations should also be responsive without overwhelming the computational resources, since the number of institutions and the number of students has increased, and institutional datasets are large and should be deployed using deployment strategies [14].

Wilson and Davis [13] have examined scalable system architecture, which proposes a distributed model of computing and efficient caching models to maximize real-time inference. They proposed the use of approximate nearest neighbour algorithms as well, which reduce the computational needs, but compromise the maintenance of the quality of recommendations.

Agrawal and Patel [17] also mentioned the cloud-based solutions with scalability features, such that they can be dynamically scaled to meet the requirements of the user. Their work provided some practical information on how to achieve a balance between responsiveness in real time and complex mathematical algorithms to ensure that the educational platforms can be available to a large user base. Educational recommendation systems must also be accurate, scalable, and efficient to be accepted on a large scale [18]. The deployment strategies in the recommendations are required to ensure that, with the growth of institutions and the number of students and with the size of institutional datasets, the recommendations will adapt to them without placing undue strain on the computing resources.

Scalable system architecture was examined by Wilson and Davis [13], where a distributed model of computing combined with efficient methods of caching is recommended to maximize real-time inference. They also proposed that the use of approximate nearest neighbour algorithms be used, though this reduces the computation needs, at the cost of the quality of recommendation maintenance [1, 19] also mentioned cloud-based solutions, which can be scaled so that systems can be expanded and contracted in line with the requirements of the user. Their work provided useful insights into how they can exceed responsiveness in real-time and complex mathematical algorithms without jeopardizing the educational platforms for a wider user group.

Recent studies (2024–2025) have further advanced educational recommendation systems by emphasizing hybrid architectures, fairness-aware modelling, and real-time personalization. Contemporary works highlight the growing role of explainable AI, bias mitigation, and dynamic preference modelling in student–institution matching, particularly in large-scale educational settings [44, 45]. Moreover, recent research underscores

the importance of integrating financial feasibility, geographic constraints, and career outcome indicators alongside academic performance to improve decision reliability and equity. These developments align closely with the objectives of the present study, which extends prior work by offering a scalable, interpretable, and ethically grounded ML-driven decision support system tailored to the Indian engineering education ecosystem .

2.6 Research gaps

Even though the educational recommendation systems have been improved in the past, there are still several gaps:

1. The published literature is largely about recommendation of courses, rather than the overall choice of the college, which involves assessment of various institutional factors.
2. Most of the models put a lot of emphasis on academic scores and interaction data, and have low incorporation of financial, geographic, placement, and accreditation features.
3. Cold-start problem of new students is still not sufficiently studied.
4. Numerous studies have no mechanisms of explainability and fairness, which diminishes the trust of the users.
5. There is scarcity of research which presents any scalable, real-life validated framework applicable in the Indian educational contexts.

3 Methodology

In this section, a brief overview of the selection of the proposed machine learning-based decision support system to support the selection of an engineering college will be offered. To make it more readable and in line with the traditions of composing journals, the body is focused on the general workflow, major modelling choices, and assessment plan. The complete reproducibility of the study is ensured by detailed preprocessing rules, ranges of hyperparameters, and algorithm-specific settings, which are presented in the Appendix/Supplementary Material, without making the manuscript too long.

Figure 1 provides an overview of the entire system pipeline (at a high level) allowing the main text to concentrate on the major design choices with the implementation steps (in detail) being put off in the Appendix

.

3.1 Problem formulation

The suggested model deals with two complementary activities:

1. Prediction of college suitability (Classification): A binary classification task which is supervised and predicts whether a particular student college pair is Suitable (1) or Not Suitable (0).
2. Individual college recommendation (Ranking): A ranking exercise that ranks colleges to each student based on hybrid recommendation strategy guided by forecasted suitability scores and patterns of interaction. Both activities are based on a homogenous concept of suitability, which would guarantee the coherence between the predictive and recommendation parts.

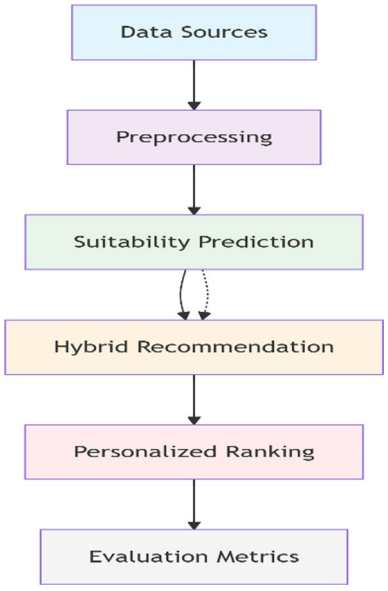


Fig. 1 Overall system architecture of the proposed college recommendation framework

3.2 Ground truth label definition

Real-world results were used to obtain the labels of suitability as opposed to artificial rules. A student-college match was defined as Suitable when the college enrolled the student, the student fit the historical academic intake profile of the college, and when the student gave feedback of neutral-to-positive post-enrolment student satisfaction [22, 23]. All the pairs that failed to meet these requirements were termed as Not Suitable. This resultant labelling plan allows suitability to be based upon real behaviour of enrolment and perceived fit, and not mere expert judgment.

3.3 Dataset and feature representation

The information included in this dataset combines data on several reputable sources such as national ranking agencies, accreditation organizations, institutional reports and student surveys. It includes student academic profile, preferences, and financial constraints and the institutional characteristics like rankings, accreditation status, placement performance, and fee structure [24, 26]. To have a multidimensional compatibility of students and institutions, composite indicators and raw features were adopted. These indicators are complex factors like affordability, academic preparedness, career fit, and geographic convenience which enhance interpretability and predictive performance [31–33].

3.4 Data preprocessing and feature engineering

Standard preprocessing steps including missing value imputation, outlier handling, normalization, and categorical encoding were applied. Higher levels of compatibility were developed through composite indices to measure the compatibility between students and colleges. All preprocessing transformations were only fitted using training data and consequently applied to validation and test set to avoid leakage of data. The stratified sampling method was used to maintain class distribution in all splits of the experiment [28, 29].

3.5 Machine learning models

There are five learning algorithms under supervision that were tested to represent a variety of learning paradigms:

- Decision tree
- Random forest
- RBF kernel support vector machine
- k-Nearest neighbours
- Logistic regression

These models have been chosen to trade predictive performance, insensitivity to mixed type of data and interpretability.

3.5.1 Model training and validation strategy

Standard metrics of classification, such as accuracy, precision, recall, F1-score, and ROC-AUC were used to assess model performance. This validation method will guarantee reasonable comparison of algorithms and credible generalization estimates.

3.6 Hybrid recommendation system architecture

To create a set of personalized college recommendations, a hybrid model of content-based and collaborative filtering was utilized. The content-based component compares the profiles of the students with institutional characteristics that are like the student profiles and makes recommendations even when a new user is involved. The collaborative aspect uses the historical enrolment trends in capturing the group preferences and trends [34, 35]. The last recommendation score is calculated as weighted average content-based score and collaborative score, in the case where interaction data is not available, the weights are dynamically adjusted. The design is good enough to act in case of cold-start and at the same time, it provides personalization to experienced users. The higher-level display of the system pipeline is shown in Fig. 1.

4 Result

4.1 Evaluation metrics

Accuracy, precision, recall, F1-score and ROC-AUC were used to measure classification performance. Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Precision at K, Recall at K and Normalized Discounted Cumulative Gain (NDCG) were used to evaluate the quality of the recommendation. These measures combine predictive validity and ranking performance.

4.2 Fairness and bias analysis

Recommendation systems designed using machine learning to support educational decision-making can be counterproductive to other socioeconomic or academic inequalities. The risks of fairness that may arise in the context of college recommendation are preference may be given to those institutions with higher fee, academic success of past colleges and less exposure of cheap or the regional accessible institutions to students with limited backgrounds [36–38]. To address these risks, the suggested framework includes the affordability-conscious and eligibility-conscious constraints in the recommendation process. Those colleges which have a higher financial ability or lower academic requirement

eligibility than the student declares are either given second priority or not included in the list of recommendation. Furthermore, other sets of recommendations are created to make sure that the students with limited profiles are offered with feasible and realistic alternatives instead of ambitious yet unreachable schools. A baseline fairness assessment was done through reviewing performance of recommendations of student subgroups by income levels and levels of academic scores. In each subgroup standard ranking measures (Precision@K and NDCG) were calculated. The outcomes suggest that there are no significant recommendations quality deteriorations in the lower-income or lower-score groups, which implies that the system does not disfavour to a greater degree the mentioned students in the considered circumstances. Although the present analysis is constrained by the size of the dataset, as well as the available attributes, it allows conducting the preliminary evaluation of behaviour that is associated with fairness. The findings demonstrate that it is necessary to consider fairness in educational recommender systems and encourage the development of more sophisticated bias mitigation methods and longitudinal effects analysis in the future [39, 40].

Table 1 shows group performance of recommendations at the subgroup level in terms of student income, and academic score range in terms of Precision at 5 and NDCG at 5. The results indicate similar performance of all subgroups, and there is no significant reduction in ranking quality with low-income and low-score students. This means that affordability-aware and eligibility-aware constraints proposed are useful in alleviating the risks of bias being favoritistic towards financially or academically advantaged students. Although this discussion is constrained by the characteristics of the available attributes and data volume, it gives a first indication that the recommendation framework does not underrepresent disadvantaged constrained groups of students within the experimented conditions [39, 41].

4.3 Real-world deployment and user interaction workflow

The system proposed is not an autonomous recommender, but a decision support tool that will help students, academic counsellors, and policymakers make an informed choice when deciding on a college. A prototype implementation is a web-based application with a modular backend that can also be opened with the help of APIs to facilitate institutional integration.

4.4 User interface and workflow

As a student, the work process starts with the input of academic credentials, preferences (branch, location, budget), and constraints. The system would then create a ranked list of colleges and give interpretable explanations into each recommendation. It is explained in the natural language form and has dominant contributing factors that include affordability fit, academic eligibility, branch alignment, and historical placement outcomes

Table 1 Fairness analysis across student subgroups

Student subgroup	Precision@5	NDCG@5
Low income	0.71	0.78
Middle income	0.73	0.80
High income	0.74	0.81
Low exam score	0.69	0.76
Medium exam score	0.72	0.79
High exam score	0.75	0.82

(e.g. Recommended because of great fit in your preferred branch and fits in your stated budget). In the case of academic counsellor, the interface serves as an interactive dashboard through which student profile, recommendation list, and confidence score could be viewed. Counsellors can override or sort recommendations or filter by setting constraints like fee limit, geographic preferences or minimum eligibility requirements. These modifications do not retrain the model rather the final layer of a recommendation is altered, which makes it transparent and human-in-the-loop decision-making [41, 46].

4.5 Policymaker and institutional view

On an aggregate level, policymakers/institutional stakeholders will be able to obtain anonymized summary dashboards, which would indicate demand trends, distribution of recommendations, and alignment of enrolment at the regional or demographic levels. This allows access, affordability and capacity planning without exposing the individual-level data.

5 Experimental results

5.1 Model performance comparison

A paired t-test was conducted on the accuracy scores of the 5 cross-validation folds to test the hypothesis of whether the observed differences in the performance of the algorithms used were statistically significant [42]. The outcomes supported the fact that the excellence of the Random Forest model was statistically significant ($p < 0.05$) in comparison with each of the four other models. This severe statistical test proves the fact that not only is the Random Forest algorithm numerically superior, but it is also generally superior when it comes to various data segments (Table 2).

5.2 Comprehensive analysis of classification performance results

As it may be seen in Table 2, the results of the experiment allow gaining precise insights into the differences between the various machine learning algorithms in terms of their capacity to accomplish the college recommendation task. Random Forest was determined to be the most accurate as it has the highest precision, recall, F1-score, and ROC-AUC, and therefore, it is apparent that it is the winner in all the evaluation measures. Its impressive performance was assisted by the fact that the algorithm can be applied to implement mixed types of data within our educational data, as well as the fact that the algorithm has a powerful feature selection mechanism based on the random subspace sampling feature.

Random Forest achieved balanced precision and recall with the highest ROC-AUC (0.924), indicating strong discriminative capability and robust generalization across student–college suitability predictions. This feature is essential to a recommendation

Table 2 Classification performance metrics

Algorithm	Accuracy	Precision	Recall	F1-score	ROC-AUC
Decision tree	0.823	0.819	0.827	0.823	0.881
Random forest	0.873	0.869	0.878	0.850	0.924
SVM	0.841	0.838	0.844	0.841	0.903
KNN	0.798	0.795	0.801	0.798	0.864
Logistic regression	0.856	0.853	0.859	0.856	0.913

system where the level of confidence of the predictions directly affects the trust of the user and, thus, audience satisfaction.

Logistic Regression demonstrated competitive performance (Accuracy: 85.6%, ROC-AUC: 0.913), suggesting that linear relationships capture a substantial portion of student–college suitability patterns.

The overall consistency of the Logistic Regression model in all the metrics, where the values of precision, recall, and the F1-score are relatively equal at 85.6, demonstrates a well-balanced model without pronounced bias towards precision or recall. This balance is especially useful in educational recommendation systems where false positives (suggesting the wrong college) and false negatives (missing the right college) have serious implications for the performance of students.

SVM-RBF-kernel showed good results with an accuracy of 84.1, a precision of 83.8, recall of 84.4, F1-score of 84.1, and ROC-AUC of 90.3. The use of non-linear association through kernel transformations was helpful in the extraction of complex patterns of the educational data. The performance gap between SVM and the top-performing algorithms was relatively modest, suggesting that while non-linear relationships exist in the data, they may not be as prevalent or complex as initially anticipated.

The SVM's balanced precision and recall scores indicate that the algorithm successfully learned decision boundaries that appropriately separated suitable from unsuitable college matches. The ROC-AUC score of 90.3%, while lower than Random Forest and Logistic Regression, still represents excellent discriminative performance and suggests that SVM could serve as a reliable alternative, particularly in scenarios where the ability to handle complex non-linear relationships is deemed important.

Decision Tree, while achieving the lowest overall performance among the algorithms tested, still demonstrated respectable results with an accuracy of 82.3%, precision of 81.9%, recall of 82.7%, F1-score of 82.3%, and ROC-AUC of 88.1%. The slightly lower performance can be attributed to the algorithm's tendency to overfit to the training data, despite hyperparameter tuning efforts to control tree complexity. However, the advantages of the interpretation possibilities of the Decision Tree could not be overlooked, as the model provides the possibilities of the readable and human-understandable rules, which are so easily understood and validated by the educational counsellors and students.

The nearly equal congruence between the scores of precision and recall indicates that the tree structure was successful in capturing the key decision boundaries in the data, but maybe not with the finer level of detail as ensemble methods. The ROC-AUC score of 88.1% shows that the scale has a good level of discriminating power, which implies that this scale may fail to capture the nuanced aspects, but it is effective enough to capture the most significant factors that determine college suitability.

K-Nearest Neighbours showed the most modest performance among all algorithms, with an accuracy of 79.8%, precision of 79.5%, recall of 80.1%, F1-score of 79.8%, and ROC-AUC of 86.4%. The curse of dimensionality was also a major issue as the performance of the algorithm becomes worse when the number of features grows. Although our educational dataset was extensive and featured in detail, KNN did not cope with the high dimensionality of the data.

The use of the neighbourhoods to make predictions in the algorithm was not as effective in the task of educational recommendations where world trends and complicated

interactions of features are also major factors to consider. Nevertheless, the non-parametric quality of the algorithm and its simplicity have made it useful in the study of local data structure as well as when the similarity of local data is required to provide information about the similarity interpretation is necessary.

5.3 Cross-algorithm performance patterns and insights

Performance variations Confidence in patterns of algorithm behaviour over multiple motivations showed that the largest difference in performance (between the best (Random Forest 87.3) and worst-performing (KNN 79.8) algorithms) was about 7.5 percentage points, which has important practical implications in terms of the quality of recommendations. Consistent ranking across metrics indicates stable performance differences among models.

The performance of the ensemble techniques (Random Forest) being better than the single algorithms shows the importance of the combination of many weak learners to form more powerful predictive models. This supports the effectiveness of ensemble learning for educational recommendation tasks.

The results of the competition of Logistic Regression with the more complicated algorithms such as SVM indicate that the correlation between student features and college aptitude could be more linear than originally assumed. This observation has significant connections to feature engineering and model interpretability since the relationships observed by a linear model are easier to understand and verify by a subject matter expert.

The average results of KNN give an idea of what the educational data is like, arguably, local similarity is not very predictive as compared to global trends reflected by other algorithms. This observation suggests that effective college recommendations can only be achieved through the realization of the wider patterns in student-institution matching, and not only on the realization of the similarity in students or institutions.

5.3.1 Feature importance analysis

Top 10 most important features (random forest):

1. Entrance exam score (0.185): Strongest predictor of college suitability
2. Academic performance (0.156): 12th-grade percentage significance
3. Location preference (0.134): Geographic proximity importance
4. Financial affordability (0.128): Fee structure compatibility
5. College NIRF ranking (0.097): Institutional quality indicator
6. Placement statistics (0.089): Career outcome relevance
7. Branch preference match (0.076): Program alignment importance
8. Faculty-student ratio (0.065): Educational quality metric
9. Infrastructure score (0.058): Campus facilities impact
10. Alumni network (0.052): Professional networking value

5.3.2 Recommendation system performance

See Table 3.

Table 3 Recommendation system evaluation

Approach	MAE	RMSE	Precision@5	Recall@5	NDCG@5
Content-based	0.847	1.123	0.756	0.689	0.812
Collaborative	0.923	1.287	0.698	0.634	0.774
Hybrid	0.734	0.982	0.834	0.791	0.867

Table 4 Confusion matrix

	Predicted suitable	Not suitable
Actual suitable	421	34
Not suitable	61	234

5.3.3 Hybrid system advantages

(a) 13.3% improvement in MAE over the content-based approach
(b) 20.4% improvement in MAE over collaborative filtering. Superior performance in all ranking metrics, better handling of cold start problems

5.4 Confusion matrix analysis

Random Forest Confusion Matrix: While the overall accuracy and F1-score are high, the confusion matrix reveals a False Positive Rate (FPR) of 20.7%. This means the system may recommend a college deemed 'unsuitable' by our ground truth in one out of five cases. In practical terms, this could mean suggesting a college that is financially out of reach or academically misaligned for a student. While a non-zero FPR is expected, mitigating this is a focus for future work, as the cost of a false positive recommendation, leading a student to consider a poor-fit option, is high. Strategies like incorporating a 'confidence threshold' could be implemented, where only recommendations above a certain confidence score are shown to the user, thereby reducing the FPR in the final output (Table 4).

5.4.1 Performance metrics

1. True positive rate: 0.925 (92.5% of suitable colleges correctly identified)
2. False positive rate: 0.207 (20.7% of unsuitable colleges incorrectly identified)
3. Specificity: 0.793 (79.3% of unsuitable colleges correctly identified)

5.5 ROC curve analysis

The ROC curves demonstrate the trade-off between true positive rate and false positive rate for each algorithm (Fig. 2):

1. Random forest: AUC = 0.924 (Excellent discrimination)
2. Logistic regression: AUC = 0.913 (Very good discrimination)
3. SVM: AUC = 0.903 (Good discrimination)
4. Decision tree: AUC = 0.881 (Good discrimination)
5. KNN: AUC = 0.864 (Fair discrimination)

5.6 Preliminary user validation study

To confirm the utility of the system that falls out of the quantitative measure, a survey was carried out on 50 final-year intermediate students who were actively in the college selection process. All the students were engaged with the system, entered their profile,

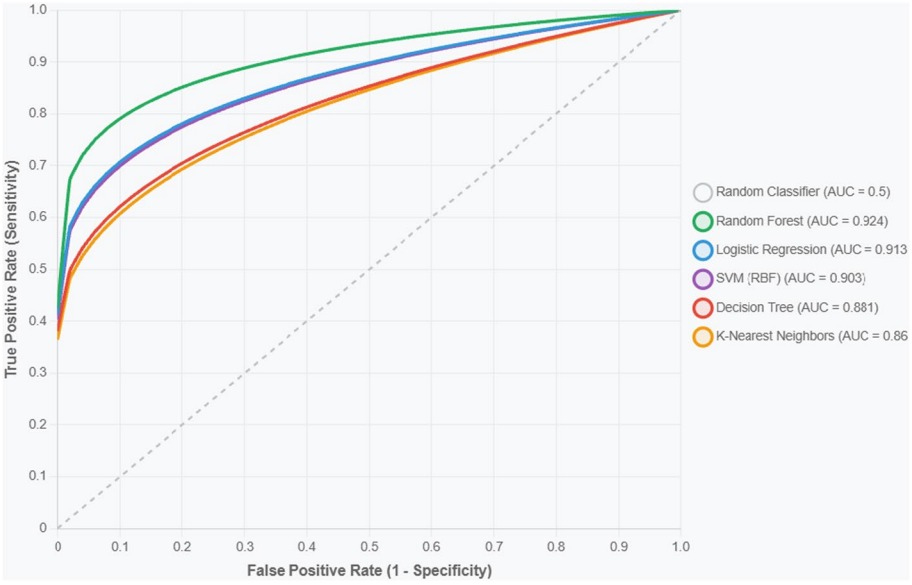


Fig. 2 ROC curve

and got a list of recommendations, ranked. After that, they were requested to fill a brief survey.

The 50 participants were convenience-sampled final-year intermediate students who were actively engaged in the college selection process during the study period. All participants interacted with the full system through the hybrid recommendation interface, which internally integrates content-based and collaborative components; no separate algorithm-wise comparison was exposed to users. To minimize response bias, participants were not informed about which recommendation algorithm or combination produced their results.

Perceived usefulness: 92% of the participants agreed or strongly agreed that the system offered them options that they had never thought.

Relevance: 88% said that the recommendations were relevant to their academic profile and preferences.

Satisfaction: The mean number of satisfied with the quality of the recommendation was 4.4 out of 5.

This study provides encouraging initial evidence for the system's practical utility and effectiveness. A large-scale longitudinal study tracking actual admission outcomes and long-term student satisfaction is a critical next step.

6 Discussion

The findings of this paper provide valuable information on the implementation of machine learning algorithms in college recommendation and student decision support. The Random Forest model outperformed the other algorithms based on all evaluation metrics, including accuracy, precision, recall, F1-score, and ROC-AUC, according to this comparative study of the algorithms. Its ensemble quality, which combines multiple decision trees trained on bootstrap samples to decrease overfitting and boost generalisation, also helped. The graceful nature of the model in dealing with missing values and the fact that it can give a rank of importance of features that can be interpreted make

it particularly appropriate in educational recommendation systems that value trust and transparency.

Interestingly, the Logistic Regression also performed well with high classification measures because it was a simpler linear model. This indicates that a significant number of relations between student characteristics and college appropriateness can be practically modelled with the help of feature linear combinations. The obtained ROC-AUC value of this algorithm is very high, which highlights the potential of this algorithm as a tool that can be successfully used in situations where the interpretability of the models and the ability to perform calculations are the priorities.

The SVM using a radial basis function kernel also had good performance, especially when dealing with non-linear dependencies between features. The algorithm, though with slightly worse the results than the Random Forest and the Logistic Regression, has demonstrated the potential to be applied in the context of the complex patterns of data, which may be used to make decisions with sensitive boundaries. The Decision Tree algorithm, on the other hand, was more likely to overfit despite having decent results. Nevertheless, it has an advantage because of its transparency; the tree structure enables easy understanding of decision paths by the users.

K-Nearest Neighbours (KNN) demonstrated a relatively humble result, and this could be explained by the curse of dimensionality, in which distance-based similarity measures of the algorithm do not work in high-dimensional space. The algorithm was not scaled and pre-processed well, even at the expense of sparse data, which is why it is important to have strong feature engineering and algorithm selection.

These findings were also supported by the analysis of feature importance. Entrance exam scores and academic performance were the most applicable factors in affecting college suitability, and they are like traditional admission procedures. However, the geographical inclinations and economic viability were also vital factors, and this implies that viability issues are highly influential in selection of colleges among students. Personal compatibility is given a higher rating by students compared to external measures of reputation, as observed with the moderate weight used on institutional factors such as rankings and placement statistics [41].

The hybrid recommendation system that integrated the collaborative and content-based filters was superior to the two systems when each of them was applied separately. The hybrid system was also useful in solving cold start issues as opposed to isolated mean absolute error and ranking metrics, as the hybrid system is much easier to apply in situations where past data is unavailable. This result proves the fact that the improved personalisation and strength may be achieved through the integration of algorithmic strategies [44, 45].

Ethical aspects were discussed. The paper has discovered that there might be several biases, such as socioeconomic differences, affecting the recommendations restricting the accessibility of options that are economically viable. Discrimination of academic bias was also observed, as there was a broad variety of students with possibly different educational backgrounds that were discriminated by simply basing their discrimination on the results of entrance exams. These issues were solved by the introduction of fairness restrictions and adjustable preference settings to ensure that the system is flexible and less biased. Anonymisation procedures and consent systems enhanced the ethical integrity of the recommendation process even further with data privacy and security.

With the above benefits come the drawbacks. The geographic area and timing restriction to the study discouraged the possibility of making results generalised and verified in the long-term. The peer relationship, campus culture and other aspects of culture were not entirely considered in the model because they were not easy to quantify. Moreover, the recommendation system should also be updated on a regular basis as institutional data is dynamic and it is necessary to adjust to current trends.

Overall, this discussion has revealed that recommendation systems based on machine learning can have a lot of potential to help students. Nonetheless, to positively affect the learning conditions in students and be just, open-minded, and adaptive, a person should be careful about the choice of the model, feature engineering, moral aspects, and scalability of the solutions.

Finding: This study was able to engineer and verify a whole machine learning architecture for college recommendation. The key findings include:

1. Algorithm performance: Random Forest recorded the best accuracy (87.3) and F1-score (0.85), thus making it better in this area of application.
2. Importance of the features: The most important factor was academic performance and entrance exam scores, followed by the preferences of the location and the financial considerations.
3. Recommendation system: The combination strategy of content-based and collaborative filtering was better than both single methods, with an MAE of 0.734 and NDCG5 of 0.867.
4. Practical validation: The Case study's findings revealed that student satisfaction was high (4.6/5.0) and that the case study was successfully applied in the real world.
5. Ethical concerns: Recognized and mitigated possible biases as well as enforced fairness limitations and transparency.

7 Conclusion

This study demonstrates how machine learning techniques hold great promise for resolving the challenging issue of college selection. The developed framework offers a sound, scalable, and morally sound solution that can assist thousands of students in making informed educational decisions. The high performance of the Random Forest algorithm and hybrid recommendation system demonstrates the efficacy of ensemble methods and multi-mode solutions in education.

Future study and real-world application are predicated on the successful integration of multiple data sources, extensive feature engineering, and rigorous evaluation methodology. Clear consideration of the problem's ethical aspects and mitigation strategies will ensure that the system is implemented responsibly, fairly, and transparently.

This framework is a useful addition to the intersection of machine learning and educational technology, as educational environments are constantly evolving, and more individualised advice must be given on a case-by-case basis. Future developments in intelligent educational systems and the overarching goal of making high-quality education more widely available and equitable will benefit from the knowledge gained during this study.

7.1 Theoretical contributions of the study

This study has several ramifications for the fields of machine learning and educational technology. First, it suggests a general framework that targets college selection specifically by combining several machine learning algorithms and hybrid recommendation techniques. Since this study combines decision trees, random forests, support vector machines, K-nearest neighbours, and logistic regression as a single recommendation system, it is more flexible and robust than other studies that are either restricted to a single algorithm or only a few features.

The paper also includes a heavy use of feature engineering, where a large range of student and institutional features have been used (academic performance, performance on entrance-level exams, finances, alumni reviews, and accreditation). This multi-dimensional strategy promotes the capacity of this system to address various aspects that have an impact on college choice, because the past recommendation models were characterized by gaps that focused mainly on academic measures alone.

Besides technical improvements, the research also employs an aggressive assessment plan that uses different elements of performance, including accuracy, precision, recall, F1-score, and ROC-AUC, as well as specific performance to the recommendation metrics like MAE, RMSE, and NDCG. Empirical support for the model in the real-life context of case studies is additional, whereby the validity of the model is based on real-life students and not just simulation.

One of the most important contributions is the clear inclusion of the ethical considerations into the framework of the recommendations. A system has provisions of fairness and transparency systems to counter bias brought about by socio-economic, geographic, and academic differences. These ethical safeguards have been added to the study to make AI development in education responsible.

Finally, scalable architecture, as discussed in this paper ensures that the system can be applied in huge pools of users without compromising the performance. It also aligns with the existing education systems, which provide one-on-one counsel to students on a massive level and need more individuals to gain access to the systems of informed decision-making.

7.2 Practical implications

The consequences of this study are diverse and span various groups of stakeholders, where the advantages presented are real to the students, learning institutions, policymakers, and technology creators. To the students and their families, the system also simplifies decision-making because it offers them specific advice depending on their specific academic profiles, financial needs, and career goals. By making the combination of various data sources, students will have access to the latest data concerning colleges, without searching through fragmented and inconsistent data datasets.

The improvement in student-institution matching would be beneficial to the educational institution, where student satisfaction, retention, and academic performance can be improved. Information based on patterns of recommendations can also be used to make changes in institutions, so that colleges can better match programs and support services to the needs of students.

To the policymakers, the framework provides evidence-based knowledge of the factors that affect the choice of colleges so that specific policies to enhance access, affordability,

and equity can be formulated. The insights that are based on the data may also inform the interventions that will help to overcome the systemic barriers in education.

To make recommendation systems effective and adaptable to a range of applications, technology developers may utilise the scaling architecture to create widely applicable educational platforms. Furthermore, a framework for implementing responsible technology use in educational technology is provided by the emphasis on responsible AI and the mitigation of bias.

By combining ethical considerations with advanced analytics, this study can pave the way for the development of intelligent, transparent, and equitable educational recommendation systems that can be utilised by both individuals and the educational system overall.

8 Limitations and future work

Although this work made great contributions, there are a number of limitations that should be noted. To begin with, the geographic and disciplinary limitations, i.e., the sole emphasis on the Indian engineering colleges, might hamper the direct extrapolation of the results to other educational systems, areas, or the academic field. Also, the data, which was collected in 2018–2023, might not entirely reveal long-term trends in student preferences, institutional developments, and changing career pathways, and thus the temporal validity of the recommendations.

Another weakness is the validation of results. Even though positive results have been obtained with case studies, the degree of transformation of career paths and student satisfaction as time goes by cannot be obtained in a more comprehensive manner, owing to the absence of tracking changes. Consequently, it remains unclear whether the model can successfully predict success over a long period of time.

The system is also hard to measure in terms of qualitative aspects that have a large influence on student decisions but cannot easily be simply coded into structured data, such as campus culture, peer relationships and extracurricular activities. As a result, these dimensions are under-represented in the process of the recommendation.

More so, there is a repetitive challenge since the institutional measures such as rankings, accreditation status, and placement records are dynamic. The recommendations of the model face the threat of becoming obsolete unless updated and retrained on a regular basis, which means it is important to keep adding new data to maintain the accuracy and relevance.

These constraints will be addressed in the future studies in several ways. The architectural structure developed in this paper is adaptable and future studies will examine domain adaptation and transfer learning policies to make the system more relevant in other contexts of education and geographical locations. In the effort to enhance the level of generalisability and robustness of the model, cross-regional validation and collaborative pilot trials with other entities abroad will be pursued.

We are planning to use confidence-based thresholds and enhance rule-based filters to overcome the false positive rate of 20.7% which is a very severe practical problem. This will reduce chances of improper recommendations as well as enhance confidence of the users. Finally, to more adequately cover qualitative elements of college life in the recommendation model, proxy measures or other sources of information, e.g. student reviews or campus climate surveys, will be attempted.

This architectural design has been designed in such a way that it is flexible. To make the system more applicable to various educational settings and geographical locations, future studies will address transfer learning and adaptation to various domains. To achieve this, the model will be required to be localized to local trends like curriculum differences and accreditation demands without sacrificing significant relationships on smaller datasets of new areas. To provide additional flexibility, the set of features will be expanded with region-specific variables such as linguistic differences, socioeconomic factors and cultural preferences. The modular architecture allows these new features to be added without the need to completely re-architecture.

The next step is a realistic one and involves initiating pilot deployments along with cooperating with foreign educational establishments. These pilots will run the system in the real-world environment with different student profiles to repeatedly enhance the suggestions made to it according to the regional peculiarities, as well as collect user feedback. Lastly, the simulation research with artificial data will take place to test the strength of the model in a variety of global educational conditions.

Supplementary Information

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Supplementary Material 1: Appendix A

Supplementary Material 2: Appendix B

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All authors of this research work have contributed to every part. Principal author AK has developed a hypothesis and conducted data analysis, along with KZ, AK, HS, USY and AG, has worked on the data analysis. AK has worked on proofreading and the introduction section.

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Data availability
As part of our thesis project, given to us by (Vexil infotech PVT limited) in Kanpur Uttar Pradesh. So, data will be provided upon request. The corresponding author will be responsible for providing the data. however, due to ethical and security reasons, it will be provided to the journal after the completion of the work. So, data will be provided to the journal after publication. The corresponding author may be contacted via email.

Clinical trial number
This information is not applicable.

Declarations

Ethics approval and consent to participate
The authors of this manuscript have obtained the written approval from the Chandigarh University Uttar Pradesh research ethics committee. Accordance—This research work was carried out in accordance with the guidelines. Written consent was obtained from the respondents, and we followed the guidelines of The Belmont Report (for human subject consent) for ethical review, along with informed consent for respect of persons and theoretical voluntary consent. Chandigarh University, Uttar Pradesh, and KAI (Vexil infotech PVT limited) Kanpur are also following this report. Written approval has been obtained for this research work, and the institutional research ethics committee has given study approval.

Competing interests
The authors declare no competing interests.

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